Effat University — College of Engineering
Department of Computer Science

CS3081 Artificial Intelligence

LAB 3: Propositional Logic & Model Checking

Group Challenge

Instructor	Dr. Naila Marir	**Duration**	1 hour 30 minutes
Semester	Fall 2026	**Group size**	3–4 students
Submission	One GitHub link per group	**Total marks**	100

Group No.			
Member name	**Student ID**	**Member name**	**Student ID**
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1. Objectives

1. Represent facts as propositional logic sentences using Not, And, Or and Implication.
2. Explain how `model_check` decides whether a knowledge base entails a query.
3. Encode a new logic puzzle in Python and solve it with model checking.

2. Tools and Files

1. Python 3 and the package `termcolor` (install with `pip install termcolor`).
2. Lab files (on Blackboard): `logic.py`, `harry.py`, `clue.py`, `puzzle.py`, `mastermind.py`. Put all of them in one folder.
3. A GitHub account for each group.

3. Lab Plan

Step	Activity	Time
0	Setup: run harry.py and clue.py	5 min
1	Task 1 — Understand the code	20 min
2	Task 2 — Change the knowledge	20 min
3	Task 3 — Knights & Knaves challenge	30 min
4	Write the PDF and push to GitHub	15 min

4. Task 1 — Understand the Code (30 marks, 20 min)

1. **harry.py:** Write the truth table for `rain`, `hagrid`, `dumbledore` (8 rows). Mark the rows where the knowledge base is true. Use the table to explain why the program prints `True`.

rain	hagrid	dumbledore	Not(rain) => hagrid	hagrid or dumbledore	Not(hagrid and dumbledore)	dumbledore	KB (All True?)
T	T	T	T	T	F	T	False
T	T	F	T	T	T	F	False
T	F	T	T	T	T	T	**True**
T	F	F	T	F	T	F	False
F	T	T	T	T	F	T	False
F	T	F	T	T	T	F	False
F	F	T	F	T	T	T	False
F	F	F	F	F	T	F	False

Why the Program Prints True

1. **Only One Valid Model:**

The knowledge base is satisfied (**True**) in exactly **Row 3**, where:

$$\{\text{rain} = \text{True},\ \text{hagrid} = \text{False},\ \text{dumbledore} = \text{True}\}$$

2. **Entailment Definition:** The function model_check(knowledge, rain) verifies whether the knowledge base entails the query ($KB \vDash \alpha$). Entailment holds if and only if in **every** model where the knowledge base is true, the query (rain) is also true.

2. **clue.py:** The program prints MsScarlet, library and knife. Explain in plain English why each one must be true.

In the game modeled by clue.py, the secret envelope holds exactly one person, one room, and one weapon, while cards dealt to players rule out potential suspects, rooms, or weapons.

Here is why each of the three must be in the envelope:

- **MsScarlet:** The culprit must be one of three people: ColMustard, ProfPlum, or MsScarlet. The known cards reveal that ColMustard is in hand (Not(mustard)) and ProfPlum is in hand (Not(plum)). Since neither of them can be the perpetrator, **MsScarlet** is the only person left.

- **knife:** The weapon used must be the knife, revolver, or wrench. The initial dealt cards confirm the revolver is in hand (Not(revolver)). Later, an opponent showed a card from the set containing MsScarlet, library, or wrench (Not(scarlet) Or Not(library) Or Not(wrench)). Since we already proved MsScarlet and the library are the solution (meaning they are in the confidential envelope and cannot be held by players), the opponent must hold the wrench (Not(wrench)). With both the revolver and wrench accounted for in players' hands, the **knife** is the only weapon remaining.

- **library:** The crime scene must be the ballroom, kitchen, or library. Known cards confirm the kitchen is in hand (Not(kitchen)) and the ballroom is in hand (Not(ballroom)). With both of those rooms eliminated, the murder must have taken place in the **library**.

5. Task 2 — Change the Knowledge (30 marks, 20 min)

For each change: **(a)** predict the result, **(b)** run the code, **(c)** explain the result.

1. **harry.py:** Delete the last fact `dumbledore` from the knowledge base. What does the program print now? Why?

The program prints **False**.

Reason: When dumbledore is removed, the knowledge base no longer forces dumbledore to be true and hagrid to be false.

Consider the possible world where:

- rain = False
- hagrid = True
- dumbledore = False

Evaluating the remaining rules in this world:

1. Not(rain) => hagrid: True => True (True)
2. hagrid or dumbledore: True or False (True)
3. Not(hagrid and dumbledore): Not(True and False) (True)

All statements in the knowledge base evaluate to True, meaning this is a valid model.

Because model_check requires rain to be true in **every** model where the knowledge base is true, having a valid model where rain is False breaks entailment, causing the function to return False.

2. **puzzle.py:** Delete the clue `MinervaGryffindor`. Who can still be placed in a house? Then add **one new clue of your own** that makes the answer unique again.

Who can still be placed in a house?

Nobody can be placed in a house. The program prints nothing.

Why: Without MinervaGryffindor, there are multiple valid assignments, so no single placement holds true in every model:

- Gilderoy can be in Gryffindor or Ravenclaw.
- Pomona cannot be in Slytherin, leaving Gryffindor, Hufflepuff, or Ravenclaw.
- Minerva and Horace have no specific constraints, leaving them open to any remaining houses.

Because model_check requires a statement to be true in all valid models, no symbol is entailed.

New Clue to Make the Answer Unique:

knowledge.add(Symbol("MinervaRavenclaw"))

Why it makes the answer unique:

1. Minerva is placed in Ravenclaw.
2. Because Ravenclaw is taken, Gilderoy must go to Gryffindor.
3. Since Gryffindor and Ravenclaw are taken, and Pomona cannot be in Slytherin, Pomona must go to Hufflepuff.
4. Horace is left with the only remaining house, Slytherin.

All four characters now have a single, unique placement.

6. Task 3 — Group Challenge: Knights & Knaves (30 marks, 30 min)

On an island, a **knight** always tells the truth and a **knave** always lies. Every person is one or the other, never both. Create a new file `knights.py` that imports `logic.py` and solves the puzzles below.

Hint — use this pattern for what a person says:

```
Implication(AKnight, statement)      # a knight's statement is true
Implication(AKnave, Not(statement))  # a knave's statement is false
```

1. **Puzzle 0:** A says "I am both a knight and a knave."
2. **Puzzle 1:** A says "We are both knaves." B says nothing.
3. **Puzzle 2:** A says "We are the same kind." B says "We are of different kinds."

Bonus (+10 marks) — Puzzle 3: A says either "I am a knight." or "I am a knave." (you don't know which). B says "A said 'I am a knave'." B also says "C is a knave." C says "A is a knight."

7. Submission

Create **one public GitHub repository** named `Lab3_GroupX` and submit **only its link** on Blackboard. The repository must contain:

1. `Lab3_GroupX.pdf` — group number, member names, and for each task: your answer, your code, and a screenshot of the output.
2. All `.py` files — `logic.py`, your changed `harry.py` and `puzzle.py`, and your new `knights.py`.
3. `README.md` — member names and one line on how to run `knights.py`.

Before you submit: open your link in a private (incognito) browser window to check that it works.

8. Grading Rubric

Task	What is checked	Marks
Task 1	Correct truth table and clear explanations	30
Task 2	Predictions, results and explanations; a working new clue	30
Task 3	Puzzles 0–2 encoded correctly with output and explanation	30
Submission	Clear PDF and complete GitHub repository	10
Total		**100**
Bonus	Puzzle 3 solved and explained	+10